

Introduction to Cisco Telemetry Broker (CTB)

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Distinguished Speaker & Chief Architect

Cisco Webex App

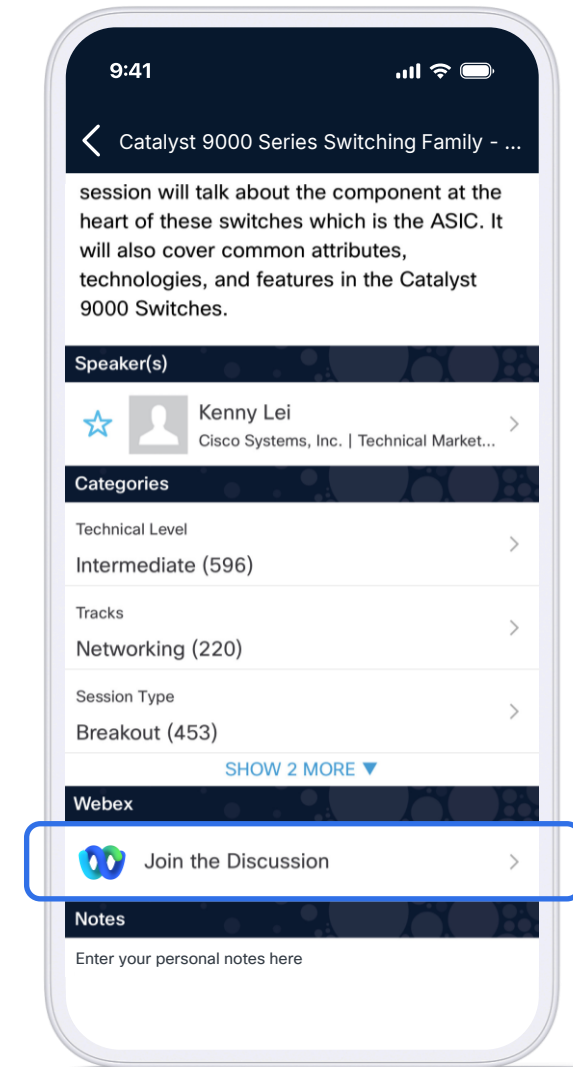
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<https://ciscolive.ciscoevents.com/clamer26/DEVNET-2340>

#> whoami



Multicast, NTP



High-speed Routing



IGMP



Network Security



IoT, ML, AI



Telemetry,
NDR



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Distinguished Engineer
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Agenda

1. **The Telemetry Conundrum**
2. **Intelligent Telemetry Plane**
3. **Deployment**
4. **Demo**
5. **Learn more**



The Telemetry Conundrum

BEFORE CTB

SOURCES



Switch



Router



Firewall



Application



IoT Device

25
FLOWS TO MANAGE

DESTINATIONS



Network Mgmt



Data Lake



SIEM / Logs



Cloud Service



Security Mgmt

AFTER CTB

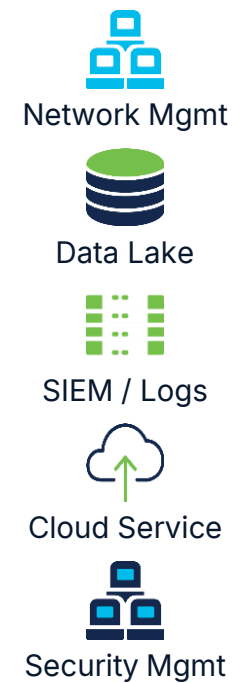
SOURCES



Cisco Telemetry Broker

BROKER · FILTER · TRANSFORM

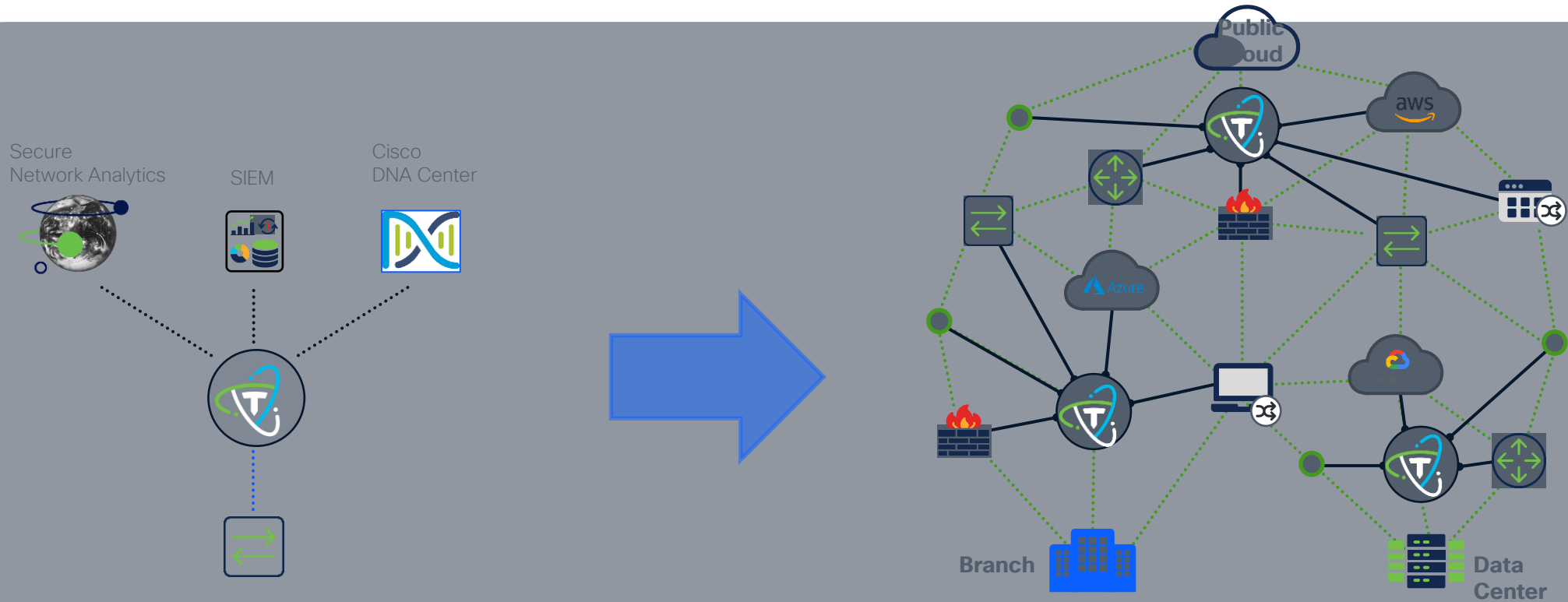
DESTINATIONS



Intelligent Telemetry Plane



Intelligent Telemetry Plane (ITP)



Control

Reliability

Availability

Visibility

Extensibility

Cisco Telemetry Broker Democratizes Telemetry Data



Brokering

The ability to route and replicate telemetry data from a source location to multiple destinations

Brokering

Replicate Telemetry

(20x perf improvement over UDP Director)



Quickly enable PoV/onboarding of non-incumbent tools

Route Telemetry

(Cloud, On Prem, Container, Serverless)



Let teams run the tools of their choice without deploying new agents/collectors



Filtering

The ability to filter data being replicated to enable fine grain control over what destinations ingest and analyze

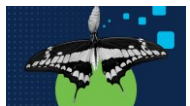
Filtering

Filter to Drop & Segment



Control Costs: Pay Splunk to only index high value data

Compliance: Keep low value data in low-cost storage



Transforming

The ability to transform data protocols from the source to the destinations protocol of choice

Transformation

Protocol Transformation



Increased visibility of legacy sources, and into cloud sources

Data Transformation



Aggregate data, transform fields, add context

Anonymize PII



GDPR compliance to all downstream tools

Telemetry Exporters

Netflow / sFlow

Syslog

App. Exporters (NVM)

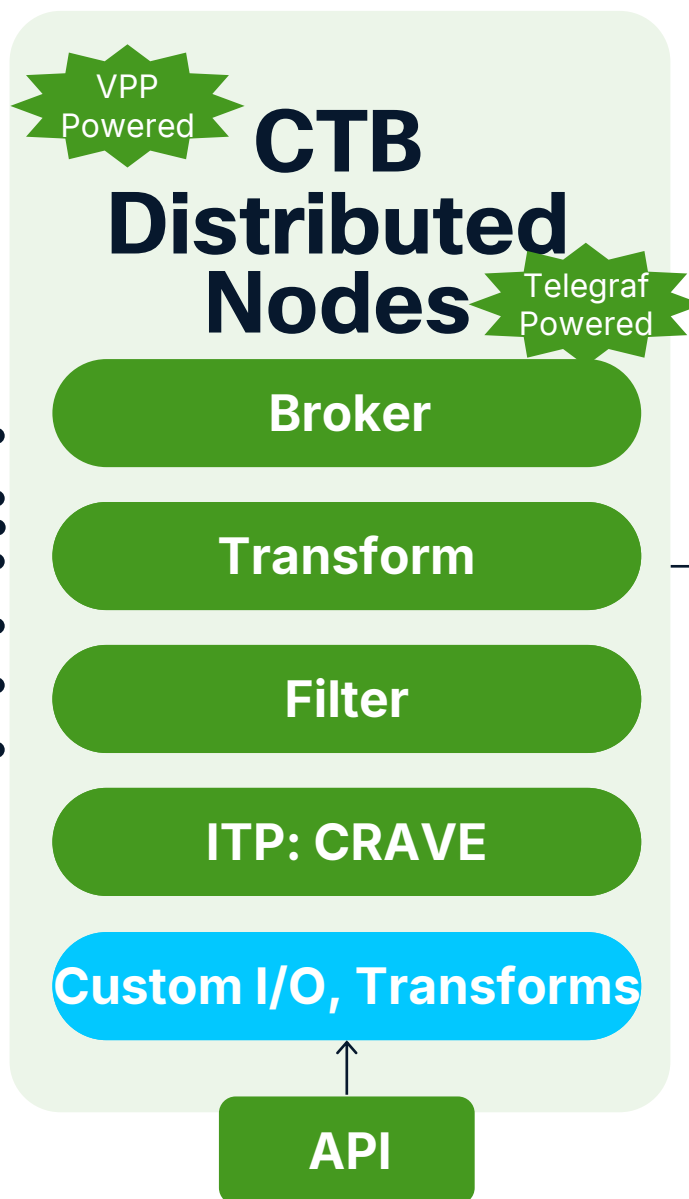
SNMP Trap

AWS/Azure Logs

Proxy logs

Generic Inputs

Exporters = Network, Application,
or Cloud provider points of
telemetry egress.



Telemetry Consumers

Cisco Secure Analytics
(SNA, SCA)

Cisco On-Prem Platforms
(DNA Center, Tetration)

3rd Party On-Prem
(Home grown Data Lakes, Live Action, SevOne)

Cisco XDR

Splunk

Generic Outputs

...and more

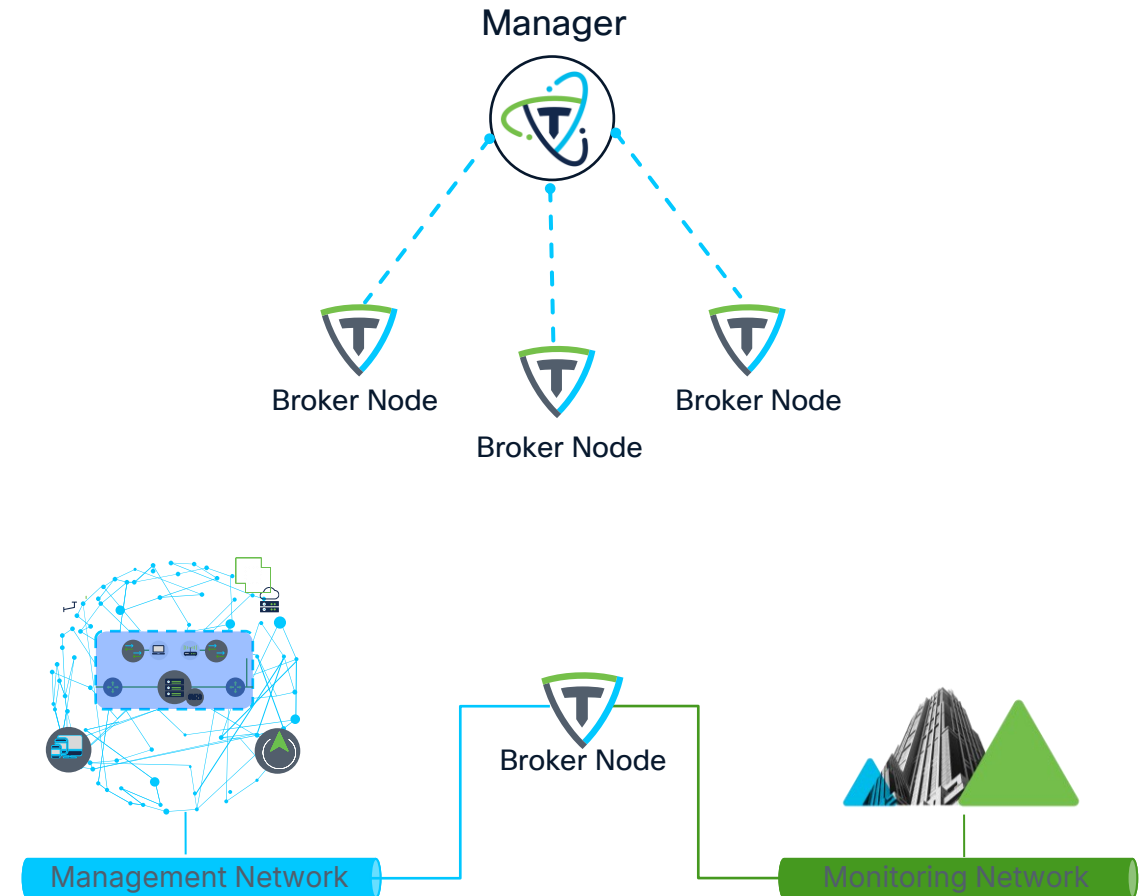


Deployment



CTB Components

- CTB Manager node:
 - Only one manager is deployed and can manage multiple Broker nodes*
 - Maintains the policy/rules for the broker nodes enabling central management from one view
 - If the manager goes down, broker nodes continue to process telemetry
 - Backup configurations are created for recovery
- CTB Broker node:
 - Where the telemetry brokering work occurs
 - Can be deployed closest to telemetry sources



Deployment specifications

Minimum requirements for a CTB Manager:

- CPU: 4 cores
- Memory: 8 GB
- Storage: 100GB



Versions
7.0, 6.7 or
6.5



Minimum requirements for a CTB Broker Node:

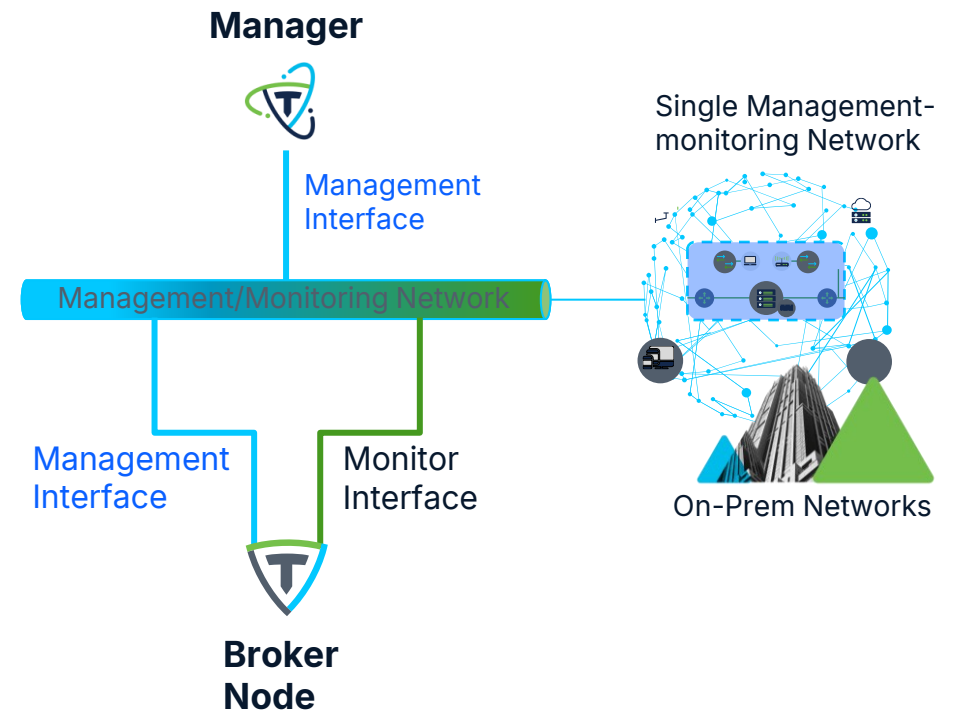
- CPU: 8 cores
- Memory: 12 GB RAM
- Storage: 60 GB



TB2300 now available (Broker Node only)
(TB2400 available FY27)

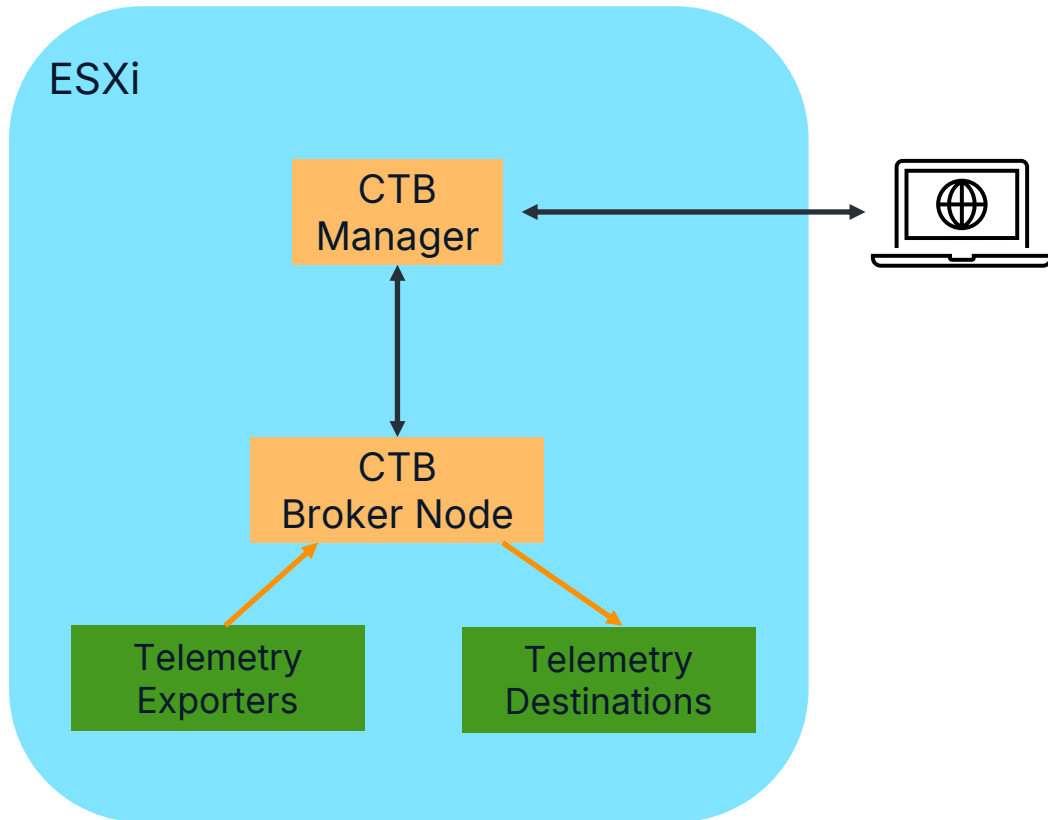
CTB Network Design (Basic)

- The Manager node has a single interface
 - Dedicated for management traffic only
- Broker nodes have two virtual interfaces
 - One dedicated for management
 - One dedicated for monitoring UDP traffic
- For smaller deployments that do not have a segmented management network all exporters of telemetry are sent to the Broker Node on a single management/monitoring network
- The Broker Node must have two IPs (v4 or v6) for Management and Monitoring interfaces (can be on same subnet)



CTB Demo

Demo Setup



- 1 Broker Node has been registered to the Manager
- 1 Telemetry exporter has been configured to send IPFIX data to Broker Node's Telemetry Interface

Demo 1.1: Configure Inputs

The screenshot displays the Cisco Telemetry Broker Explorer web interface. The top navigation bar includes the Cisco logo, the title "Telemetry Broker", and a user profile for "admin". The main interface is divided into a left sidebar and a central content area. The sidebar contains a "Dashboard" button and a "Settings" button. The central content area is titled "Explorer" and shows a breadcrumb path: "Manager Node > tb-233-devnet". Below this, there are tabs for "Data Flow", "Inputs 1", "Outputs 0", and "Broker Node Details". The "Inputs 1" tab is active, showing a list of inputs. The first input is "Cisco Live UDP Input", which is currently at "0 GB" and has a warning icon. To the right of the inputs, there is a section for "Outputs 0" with an "Add Output" button. Below the outputs section, there is a large purple information icon and a text block that reads: "Add Output. Start adding outputs by clicking the 'Add Output' button. Outputs will appear here. You can export your current UDP Director destination and rules configuration as an XML file." At the bottom of this text block is a button labeled "Upload XML file".

Demo 1.2: Configure Destinations

The screenshot displays the Cisco Telemetry Broker Explorer interface. The top navigation bar includes the Cisco logo, the text "Telemetry Broker", and user controls (help, notifications, and a user profile labeled "admin"). The left sidebar contains a "Dashboard" icon and two menu items: "Explorer" (selected) and "Settings". The main content area is titled "Explorer" and shows a breadcrumb path "Manager Node > tb-233-devnet". Below this, there are four tabs: "Data Flow" (active), "Inputs 1", "Outputs 1", and "Broker Node Details". The "Data Flow" tab is divided into two sections: "Inputs 1" and "Outputs 1". Each section has a search bar, a filter icon, and a dropdown menu (set to "Most Received Last 24h" for inputs and "Most Recently Added" for outputs). Below these sections, there are two cards. The left card, titled "Cisco Live UDP Input", shows a status of "0 GB" and a warning icon. The right card, titled "Cisco Live Collector", also shows a status of "0 GB".

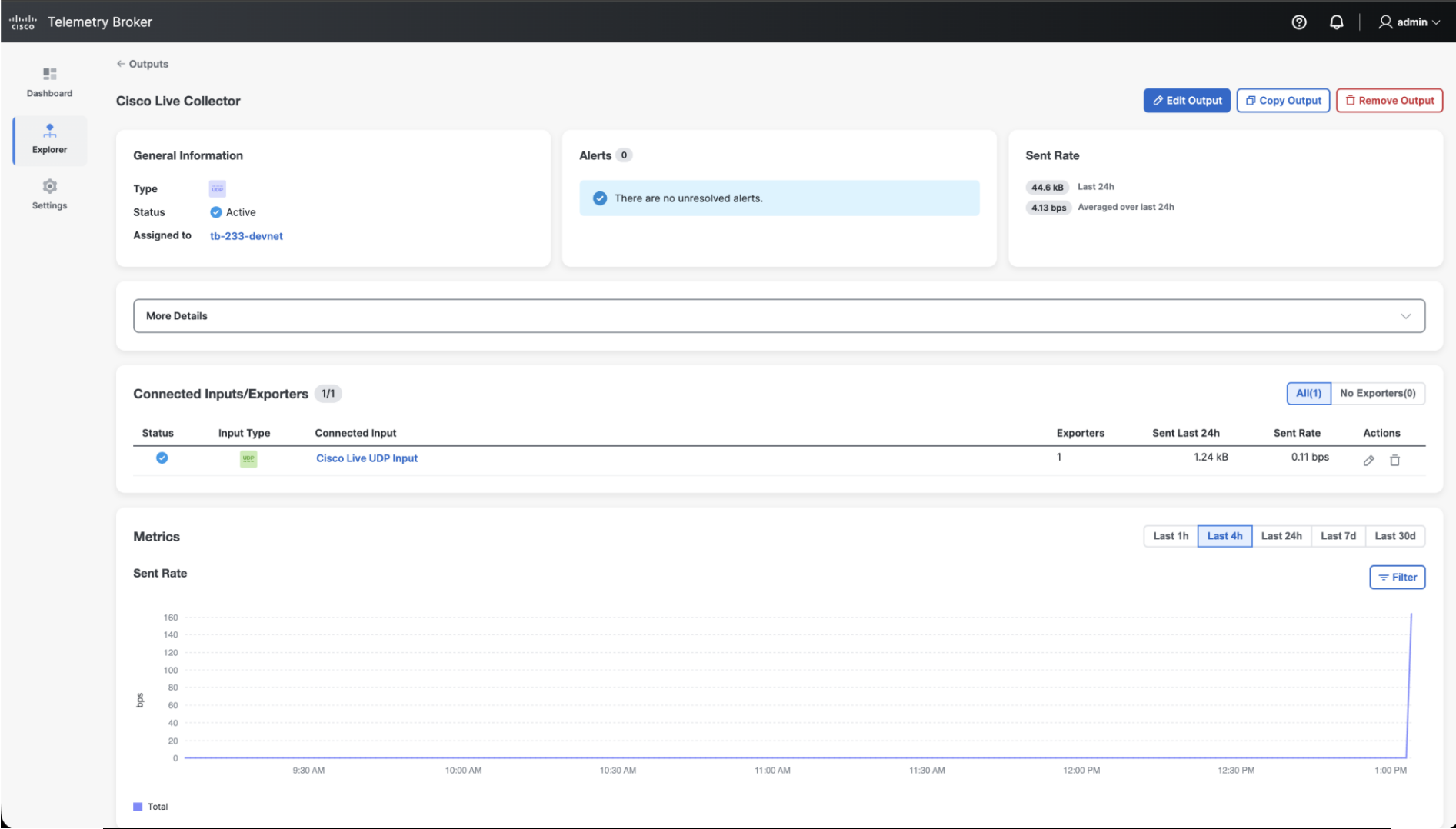
Demo 1.3: Configure Rules

The screenshot displays the Cisco Telemetry Broker Explorer interface. The top navigation bar includes the Cisco logo, the text "Telemetry Broker", and user information for "admin". The left sidebar contains a menu with "Dashboard", "Explorer" (selected), and "Settings". The main content area is titled "Explorer" and shows a tree view of "Manager Node" with a sub-entry "tb-233-devnet". The breadcrumb path is "Manager Node > tb-233-devnet". Below this, there are tabs for "Data Flow", "Inputs 1", "Outputs 1", and "Broker Node Details". The "Data Flow" tab is active, showing a visual representation of a data flow rule. On the left, under "Inputs 1", there is a search bar, a filter icon, and a dropdown menu set to "Most Received Last 24h". Below this, a card for "Cisco Live UDP Input" is shown with a "0 GB" status. On the right, under "Outputs 1", there is a search bar, a filter icon, and a dropdown menu set to "Most Recently Added". Below this, a card for "Cisco Live Collector" is shown with a "0 GB" status. A blue line with a "+" connector links the "Cisco Live UDP Input" card to the "Cisco Live Collector" card, indicating a data flow rule configuration.

Demo 1.4: Visualize Data

The screenshot displays the Cisco Telemetry Broker Explorer interface. The top navigation bar includes the Cisco logo, the text "Telemetry Broker", and user controls (help, notifications, and a user profile labeled "admin"). The left sidebar contains navigation options: "Dashboard", "Explorer" (selected), and "Settings". The main content area is titled "Explorer" and shows a tree view under "Manager Node" with a sub-entry "tb-233-devnet". The breadcrumb path is "Manager Node > tb-233-devnet". Below this, there are tabs for "Data Flow" (selected), "Inputs 1", "Outputs 1", and "Broker Node Details". The "Data Flow" tab shows a visual representation of the data flow. On the left, under "Inputs 1", there is a search bar and a dropdown menu set to "Most Recently Added". Below this, a card for "Cisco Live UDP Input" is shown with a status icon (green checkmark), a size of "22.9 kB", and a count of "1". On the right, under "Outputs 1", there is also a search bar and a dropdown menu set to "Most Recently Added". Below this, a card for "Cisco Live Collector" is shown with a status icon (green checkmark), a size of "1.24 kB", and a count of "1". A blue line connects the "1" in the input card to the "1" in the output card, indicating the data flow path.

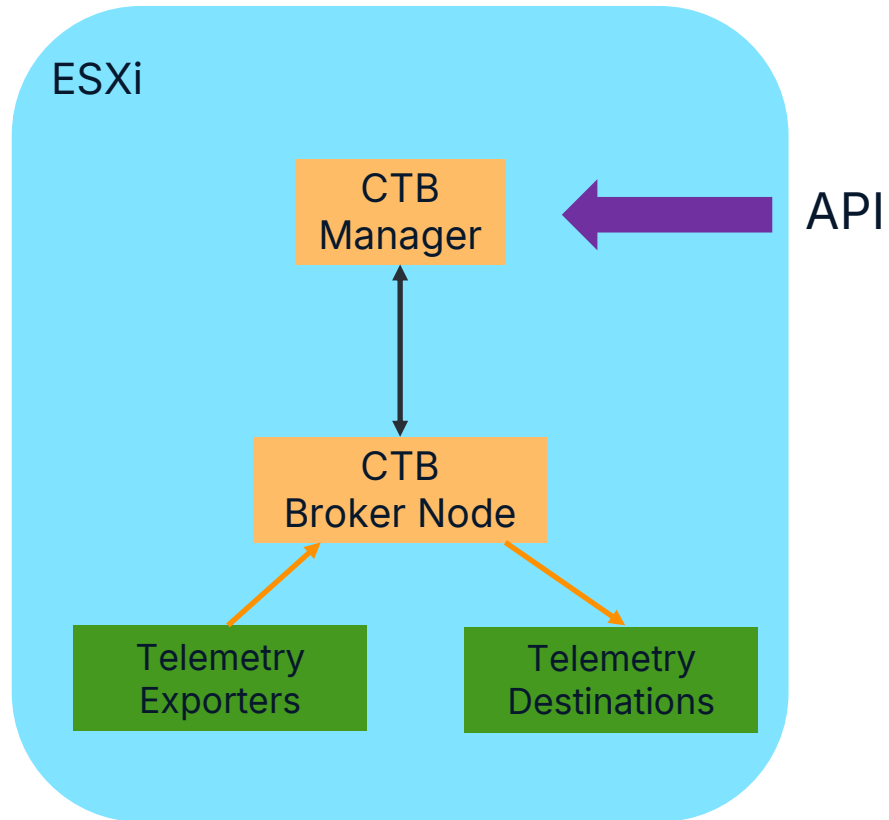
Demo 1.4: Visualize Data



CTB API Demo

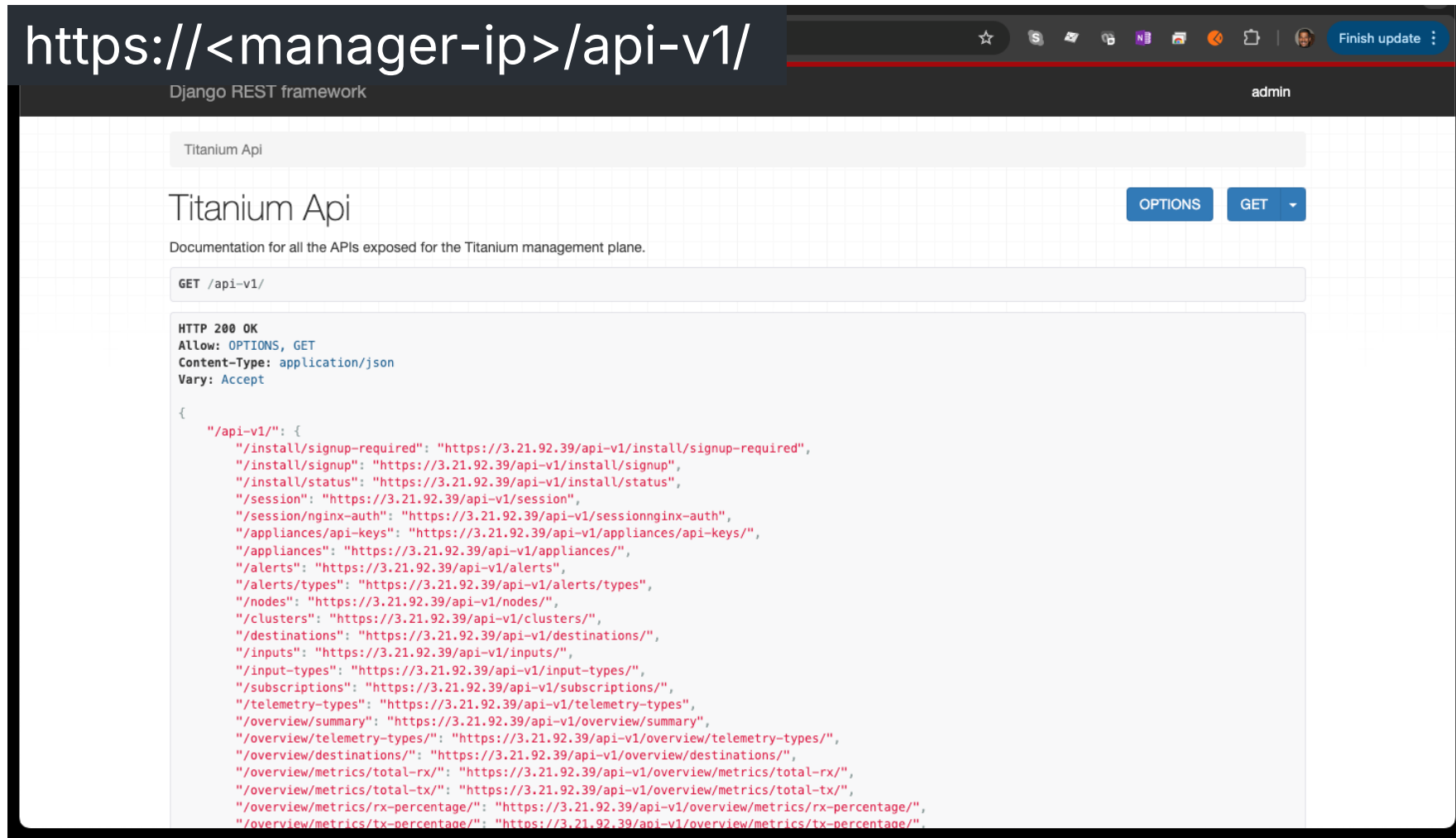
The background features a light gray gradient. A white rectangle is tilted diagonally from the top right towards the center, containing a smooth gradient from red to orange to yellow to blue. In the bottom left corner, there is a blue arc.

API Demo Setup



- CTB Manager exposes comprehensive API
- Can control all aspects of CTB via API
- Broker Nodes controlled via Manager API
- Retrieve all operational statistics via API

Demo 2.1: CTB Manager API

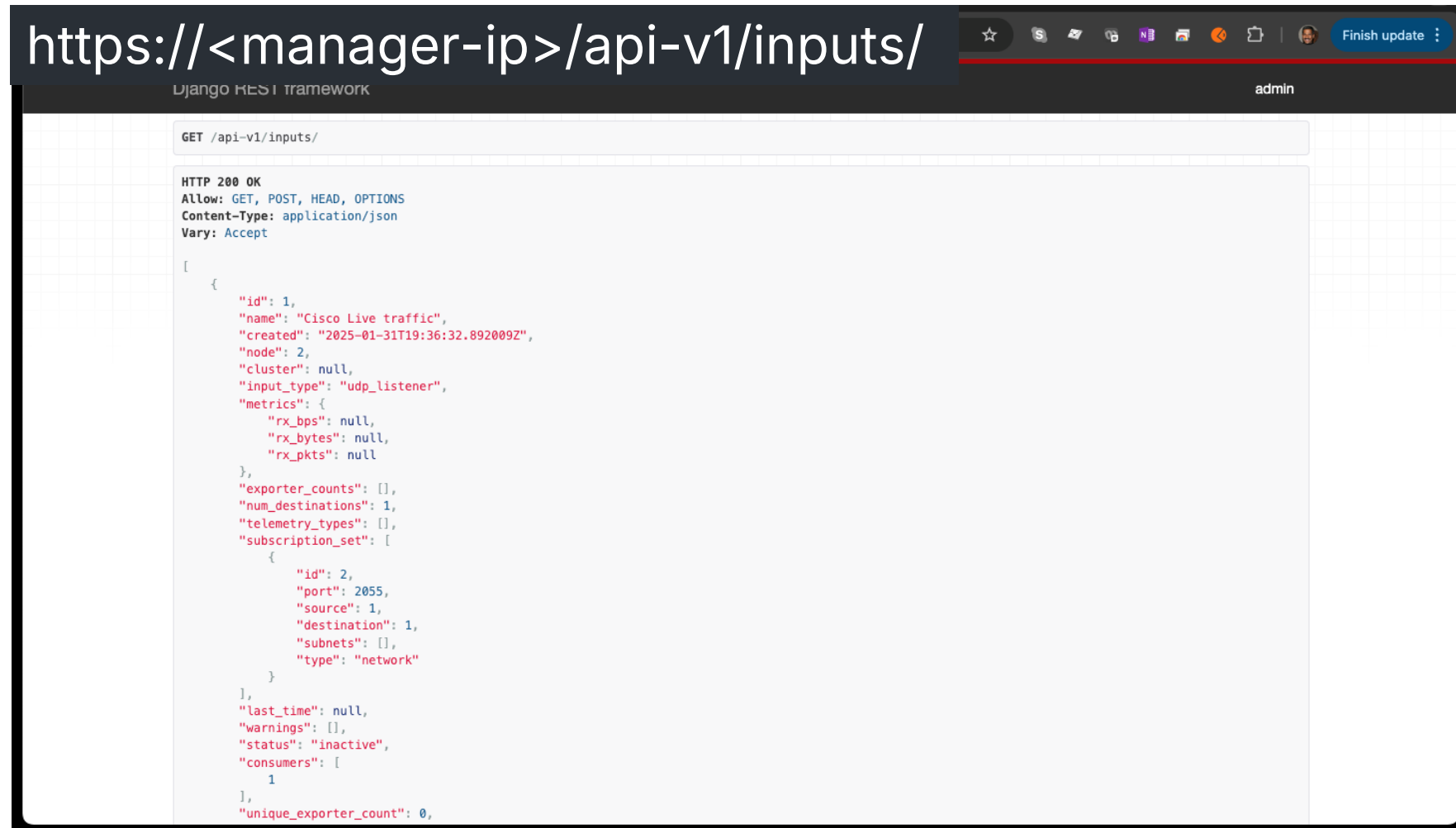


The screenshot displays a web browser window with the URL `https://<manager-ip>/api-v1/`. The browser's address bar shows a star icon, a search icon, and a "Finish update" button. The page header includes "Django REST framework" on the left and "admin" on the right. The main content area is titled "Titanium Api" and includes a description: "Documentation for all the APIs exposed for the Titanium management plane." Below the title, there are two buttons: "OPTIONS" and "GET". The "GET /api-v1/" endpoint is selected, showing the following response:

```
HTTP 200 OK
Allow: OPTIONS, GET
Content-Type: application/json
Vary: Accept

{
  "/api-v1/": {
    "/install/signup-required": "https://3.21.92.39/api-v1/install/signup-required",
    "/install/signup": "https://3.21.92.39/api-v1/install/signup",
    "/install/status": "https://3.21.92.39/api-v1/install/status",
    "/session": "https://3.21.92.39/api-v1/session",
    "/session/nginx-auth": "https://3.21.92.39/api-v1/sessionnginx-auth",
    "/appliances/api-keys": "https://3.21.92.39/api-v1/appliances/api-keys/",
    "/appliances": "https://3.21.92.39/api-v1/appliances/",
    "/alerts": "https://3.21.92.39/api-v1/alerts",
    "/alerts/types": "https://3.21.92.39/api-v1/alerts/types",
    "/nodes": "https://3.21.92.39/api-v1/nodes/",
    "/clusters": "https://3.21.92.39/api-v1/clusters/",
    "/destinations": "https://3.21.92.39/api-v1/destinations/",
    "/inputs": "https://3.21.92.39/api-v1/inputs/",
    "/input-types": "https://3.21.92.39/api-v1/input-types/",
    "/subscriptions": "https://3.21.92.39/api-v1/subscriptions/",
    "/telemetry-types": "https://3.21.92.39/api-v1/telemetry-types",
    "/overview/summary": "https://3.21.92.39/api-v1/overview/summary",
    "/overview/telemetry-types": "https://3.21.92.39/api-v1/overview/telemetry-types/",
    "/overview/destinations": "https://3.21.92.39/api-v1/overview/destinations/",
    "/overview/metrics/total-rx": "https://3.21.92.39/api-v1/overview/metrics/total-rx/",
    "/overview/metrics/total-tx": "https://3.21.92.39/api-v1/overview/metrics/total-tx/",
    "/overview/metrics/rx-percentage": "https://3.21.92.39/api-v1/overview/metrics/rx-percentage/",
    "/overview/metrics/tx-percentage": "https://3.21.92.39/api-v1/overview/metrics/tx-percentage/"
  }
}
```

Demo 2.1: CTB Manager API



```
https://<manager-ip>/api-v1/inputs/

Django REST framework admin

GET /api-v1/inputs/

HTTP 200 OK
Allow: GET, POST, HEAD, OPTIONS
Content-Type: application/json
Vary: Accept

[
  {
    "id": 1,
    "name": "Cisco Live traffic",
    "created": "2025-01-31T19:36:32.892009Z",
    "node": 2,
    "cluster": null,
    "input_type": "udp_listener",
    "metrics": {
      "rx_bps": null,
      "rx_bytes": null,
      "rx_pkts": null
    },
    "exporter_counts": [],
    "num_destinations": 1,
    "telemetry_types": [],
    "subscription_set": [
      {
        "id": 2,
        "port": 2055,
        "source": 1,
        "destination": 1,
        "subnets": [],
        "type": "network"
      }
    ],
    "last_time": null,
    "warnings": [],
    "status": "inactive",
    "consumers": [
      1
    ],
    "unique_exporter_count": 0,
  }
]
```

CTB MCP Server





Learn More



CTB Resources

- Learn more about CTB at <https://cs.co/telemetrybroker>
- Download CTB images at <https://software.cisco.com> (will need to register)
 - Deployment and User Guides have all the relevant information on how to operate CTB
- PDF of the Session will be available in the Mobile app as soon as this session is over
- If you want to try out the APIs on your CTB install, here is a link to how you can do it:
<https://github.com/brookmoor/clus2026-devnet-2340>
- Use Webex teams room for this session to ask any questions:
 - [DEVNET-2340 Introducing the Cisco Telemetry Broker \(CTB\)](#)
- Reach out to me with any questions or suggestions (<mailto:ajthyaga@cisco.com>)

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Thank you



